Environmental Economics

ECON3700

Tutorial 4 QUESTIONS

1. Table 1 below contains information on the net benefits of two projects over 4 years.

Table 1: Stream of net benefits for two projects

	Net benefits ($) in Year:				
	1	2	3	4	Total
Project A	20	20	20	20	80
Project B	50	10	10	10	80

a. What is the present value of each project assuming a 6% discount rate?

[HINT 1: $PV = \sum_{t=1}^{T} \frac{B_t}{(1+r)^t}$]

[HINT 2: $(1 + 0.06)^2 = 1.124$]

[HINT 3: $(1 + 0.06)^3 = 1.191$]

Project A:

$$PV_A = \frac{\$20}{1 + 0.06} + \frac{\$20}{(1 + 0.06)^2} + \frac{\$20}{(1 + 0.06)^3} + \frac{\$20}{(1 + 0.06)^4}$$
$$= \frac{\$20}{1.06} + \frac{\$20}{1.1236} + \frac{\$20}{1.191} + \frac{\$20}{1.262}$$
$$= 69.30$$

Project B:

$$PV_B = \frac{\$10}{1 + 0.06} + \frac{\$10}{(1 + 0.06)^2} + \frac{\$10}{(1 + 0.06)^3} + \frac{\$10}{(1 + 0.06)^4}$$
$$= \frac{\$50}{1.06} + \frac{\$10}{1.1236} + \frac{\$10}{1.191} + \frac{\$10}{1.262}$$
$$= 72.39$$

b. From (a) which project has a higher net present value? Does you answer change if the discount rate is higher or lower, why/why not?

Project B has a higher net present value. For any positive discount rate Project B will always be preferred because the net benefits occur sooner than Project A and the undiscounted net benefits are the same.

2. Table 2 reports likely benefits of costs of three programs on three people with different incomes. Which of the programs is:

a. Regressive *Program 2: Net benefits (difference between benefits and cost) as a percentage of income are highest for Person C (5%) who has the highest income, and lowest for Person A with the lowest income (1%).*

b. Progressive *Program 3: Net benefits (difference between benefits and cost) as a percentage of income are highest for Person A (10%) who has the lowest income, and lowest for Person C with the highest income (3%).*

c. Proportional *Program 1: Net benefits (difference between benefits and cost) are equal to 1% of income for A,B and C.*

Table 2: Distributional Impacts of Projects

	Person A		Person B		Person C	
Income	5,000		20,000		50,000	
Program 1						
Benefits	150	(3.0)	300	(1.5)	600	(1.2)
Costs	100	(2.0)	100	(0.5)	100	(0.2)
Difference	50	(1.0)	200	(1.0)	500	(1.0)
Program 2						
Benefits	150	(3.0)	1,400	(7.0)	5,500	(11.0)
Costs	100	(2.0)	800	(4.0)	3,000	(6.0)
Difference	50	(1.0)	600	(3.0)	2,500	(5.0)
Program 3						
Benefits	700	(14.0)	2,200	(11.0)	3,000	(6.0)
Costs	200	(4.0)	1,000	(5.0)	1,500	(3.0)
Difference	500	(10.0)	1,200	(6.0)	1,500	(3.0)

3. Name a revealed preference method for estimating the benefits of environmental policies and discuss one limitation of this method.

We discussed three methods in Lecture 4:

(a) *Travel Cost*

(b) *Defensive expenditure*

(c) *Hedonic analysis*

Each method differs in the data required for it to be used. A common limitation of these methods is they can only be used to estimate use values, and they often estimate a lower bound of willingness-to-pay for environmental quality.